

Analysis

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TPOT with constrained search space versus ensemble

2000 total evals

Version 1

Stats Testing

```
## # A tibble: 26 x 9
##   task_id .y. group1      group2  n1  n2 statistic      p p.signif
##   <int> <chr> <chr>      <chr> <int> <int>    <dbl>    <dbl> <chr>
## 1   2073 auroc ensemble_random_~ indiv~ 21  21    215  1.61e-4 ***
## 2  146818 auroc ensemble_random_~ indiv~ 21  21    216.  5.43e-4 ***
## 3  146820 auroc ensemble_random_~ indiv~ 21  21    192  6.28e-3 **
## 4  168350 auroc ensemble_random_~ indiv~ 21  21    228  4.77e-6 ****
## 5  168757 auroc ensemble_random_~ indiv~ 21  21    199  2.48e-3 **
## 6  168784 auroc ensemble_random_~ indiv~ 21  21    231  9.54e-7 ****
## 7  168911 auroc ensemble_random_~ indiv~ 20  20    194  3.22e-4 ***
## 8  190137 auroc ensemble_random_~ indiv~ 21  21    228  4.77e-6 ****
## 9  190146 auroc ensemble_random_~ indiv~ 21  21    146  3.04e-1 ns
## 10 190411 auroc ensemble_random_~ indiv~ 21  21    213  2.41e-4 ***
## 11 359955 auroc ensemble_random_~ indiv~ 21  21    196  5.42e-3 **
## 12 359956 auroc ensemble_random_~ indiv~ 21  21    179  2.63e-2 *
## 13 359958 auroc ensemble_random_~ indiv~ 21  21    231  9.54e-7 ****
## 14 359959 auroc ensemble_random_~ indiv~ 21  21    216  1.31e-4 ***
## 15 359960 auroc ensemble_random_~ indiv~ 21  21     23  5.29e-1 ns
## 16 359962 auroc ensemble_random_~ indiv~ 21  21    190  8.01e-3 **
## 17 359963 auroc ensemble_random_~ indiv~ 21  21    220  5.25e-5 ****
## 18 359964 auroc ensemble_random_~ indiv~ 20  20    158  4.84e-2 *
## 19 359965 auroc ensemble_random_~ indiv~ 20  20    153  2.07e-2 *
## 20 359968 auroc ensemble_random_~ indiv~ 21  21    192  6.28e-3 **
## 21 359969 auroc ensemble_random_~ indiv~ 21  21    231  9.54e-7 ****
## 22 359970 auroc ensemble_random_~ indiv~ 21  21    231  9.54e-7 ****
## 23 359971 auroc ensemble_random_~ indiv~ 21  21    231  9.54e-7 ****
## 24 359972 auroc ensemble_random_~ indiv~ 21  21    223  2.38e-5 ****
## 25 359974 auroc ensemble_random_~ indiv~ 21  21    230  1.91e-6 ****
## 26 359975 auroc ensemble_random_~ indiv~ 21  21    162.  1.06e-1 ns
```

Medians

```
## # A tibble: 26 x 3
##   task_id individual_tpot ensemble_random_100
##   <int>          <dbl>          <dbl>
## 1   2073          0.878          0.886
## 2  146818          0.931          0.938
## 3  146820          0.995          0.996
## 4  168350          0.961          0.965
## 5  168757          0.795          0.800
## 6  168784          0.965          0.968
## 7  168911          0.869          0.871
## 8  190137          0.916          0.926
## 9  190146          0.966          0.966
## 10 190411          0.918          0.921
## 11 359955          0.731          0.746
## 12 359956          0.934          0.934
## 13 359958          0.935          0.947
## 14 359959          0.751          0.755
## 15 359960          1          1
## 16 359962          0.843          0.853
## 17 359963          0.995          0.996
## 18 359964          0.993          0.994
## 19 359965          1.00          1.00
```

##	20	359968	0.919	0.923
##	21	359969	0.845	0.851
##	22	359970	0.906	0.925
##	23	359971	0.997	0.997
##	24	359972	0.990	0.993
##	25	359974	0.818	0.868
##	26	359975	0.994	0.995